

MISY 5360

Assignment 1

Total Points: 22

OLTP VS OLAP

Business Case

Robert Jones is a manager of several Sales Organizations at Global Bike Inc. His responsibilities are to monitor and manage sales activities. He has a number of systems to assist with the recording of day-to-day transactions, and at the end of each month he is provided with a report which displays each sale. The format of the report is illustrated below. Although this report provides a lot of information, the information is not in a format that can easily assist in the type of decisions he is required to make. The sales data is delivered by the IT Department in the following format:

YEAR	MONTH	DAY	Customer	CustomerDescr	City	Salesorg	Country	OrderNumbr	OrderItem	Product	ProductDesc	Product C	Division	SalesC	UnitOf	Revenue	Currency	Discount	CostOfGoods	Revenue USD	Discount	USD	Cogm	USD
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	10	DXTR2000	Deluxe Touri	TOU	BI		2	ST	6000	USD	300	2800	6000	300	2800	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	20	PRTR2000	Professional	TOU	BI		5	ST	16000	USD	800	7500	16000	800	7500	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	30	ORMN1000	Men's Off Rc	ORB	BI		1	ST	2400	USD	120	1200	2400	120	1200	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	40	ORHT1000	Men's Off Rc	ORB	BI		1	ST	1600	USD	80	900	1600	80	900	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	50	DXRD1000	Deluxe Road	ROB	BI		7	ST	11900	USD	595	7000	11900	595	7000	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	60	DXRD2000	Deluxe Road	ROB	BI		2	ST	3300	USD	165	1900	3300	165	1900	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	70	PRRD1000	Professional	ROB	BI		2	ST	8000	USD	400	4400	8000	400	4400	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	80	OHMT1000	Off Road Hel	ACC	AS		1	ST	50	USD	2,5	25	50	2,5	25	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	90	CAGE1000	Water Bottle	ACC	AS		1	ST	18	USD	0,9	9	18	0,9	9	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	100	RKIT1000	Repair Kit	ACC	AS		1	ST	32	USD	1,6	16	32	1,6	16	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	110	PUMP1000	Air Pump	ACC	AS		13	ST	364	USD	18,2	182	364	18,2	182	
2007	1	1	5000	Beantown Bikes	Boston	UE00	US	100001	120	FAID1000	First Aid Kit	ACC	AS		7	ST	280	USD	14	140	280	14	140	
2007	1	1	15000	Bavaria Bikes	München	DS00	DE	100002	10	DXTR1000	Deluxe Touri	TOU	BI		2	ST	4901,6	EUR	245,08	3770,2	6600	330	2800	
2007	1	1	15000	Bavaria Bikes	München	DS00	DE	100002	20	DXTR2000	Deluxe Touri	TOU	BI		1	ST	2450,8	EUR	122,54	1885,1	3300	165	1400	
2007	1	1	15000	Bavaria Bikes	München	DS00	DE	100002	30	PRTR2000	Professional	TOU	BI		3	ST	7842,54	EUR	392,13	6059,25	10559,98	528	4500	
2007	1	1	15000	Bavaria Bikes	München	DS00	DE	100002	40	PRTR3000	Professional	TOU	BI		2	ST	5228,36	EUR	261,42	4039,5	7039,99	352	3000	

The primary purpose of an information system is to process information to produce **reports to facilitate decision making**. Reports may appear in various formats and used to support a diverse range of organizational decisions. Reports provide a mechanism for organizing, analyzing, presenting and delivering information to end users. A common classification of reports is based on the types of systems which they are built from. On-Line Transaction Processing (OLTP) systems as the name suggests are **optimized for transaction processing**. They process real time information and are accessed by many users. The reports are derived from the various business transactions and predominately support tactical decision making.

An alternative information system is On-Line Analytical Processing (OLAP). This type of processing allows users to analyze information by creating **multidimensional reports**. They deal with large volumes of aggregated historical data. OLAP based reports are more flexible than the more traditional reports produced by an OLTP system.

As mentioned previously OLTP reports provide information about **particular transactions**. The flexibility of OLAP reporting assists end users in **understanding why** particular business events have occurred and or **forecast** what may occur in the future. OLAP systems and their

ability for multi-dimensional reporting are considered an important component of Business Intelligence. OLTP systems often provide the **transactional data which is used as an input for OLAP system's multi-dimensional reports**.

Your task is to suggest to Robert Jones what type of systems (OLTP or OLAP) he can use to find answers to these questions:

#	Process	Systems (OLTP or OLAP)
1	Who purchased a particular product?	OLTP
2	How much did an employee get paid?	OLTP
3	<u>How many of a product was manufactured?</u>	OLTP
4	What are the total sales for each product?	OLTP
5	What are the total sales for each department?	OLAP
6	Which salesperson has sold the most?	OLAP
7	Which products does each salesperson sell the most of?	OLAP
8	In which month did most of the sales occur?	OLAP
9	Which customer spent the least amount of money in February?	OLAP
10	Which customer spent the most in February and which sales person was responsible for the majority of these sales?	OLAP

